
WHAT DOES MULTI-AGENT LLM DEBATE ACTUALLY CHANGE?

A LAYERED ANALYSIS OF DISAGREEMENT AND ANSWER QUALITY

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ABSTRACT

Multi-agent debate, in which several LLMs exchange arguments before producing an answer, is widely assumed to improve answer quality by surfacing genuine disagreement. That disagreement is hard to verify, and no single signal can settle it, so we organize the analysis around four questions: (A) does the debater *say* it disagrees; (B) does its reply text actually *argue*; (C) does the dissent *survive* once the tone instruction that produced it is removed; and (D) do the probabilities assigned to stance options change? We evaluate three-model committees on 50 open-ended GlobalOpinionQA questions under three debate tones, *friendly* (seek common ground), *neutral*, and *hostile* (stress-test every position). The answers differ. (A) Debaters do say they disagree, and tone controls how much: full agreement differs by 50.4 percentage points between the friendly and hostile endpoints, pooling replies across all three rounds. (B) The text argues too: an LLM evaluator reading the contribution and reply without the structured self-report or tone instruction confirms the pushback is real. (C) Removing the instruction produces more returns to agreement in our sample, but the primary question-level test is inconclusive. (D) In a separate open-weight study, adjusted movement toward the opposing side is detected in two of three models, relative to a filler-based reference. This measures contextual response probabilities, not lasting belief change. For final answers, debate brings no measurable quality gain: an evaluator that judges each pair in both answer orders scores the debated answer no better than the same committee’s no-debate answer on all 299 pairs, a test that detects severe but misses moderate damage in our checks. An earlier evaluator had favored debate 66% of the time, but both its rubric and order handling differed, so this contrast does not isolate order bias. Taken together, LLM debate readily changes what agents say, but we find much weaker evidence that it changes what they persistently endorse or improves the quality of the final answer.

1 Introduction

Multi-agent LLM debate brings several models together to exchange arguments before producing a final answer. The motivation is that models can challenge one another’s reasoning and use the discussion to improve their answers. However, models may also agree through sycophancy or conformity [Sharma et al., 2023, Perez et al., 2022, Zhu et al., 2024], and weaker models may yield to more assertive peers [Xiong et al., 2023]. On tasks with known answers, debate does not consistently outperform a single well-prompted model or a matched sample-and-vote baseline [Smit et al., 2023, Li et al., 2024a, Wang et al., 2024b, Kenton et al., 2024]. These findings raise a broader question: what does the exchange of arguments actually change? On *open-ended* opinion questions, answering this requires more than checking whether models agree, because agreement alone tells us neither how their positions changed nor whether the final answer improved. We investigate these questions through a layered analysis of reported agreement, reply text, persistence after removing the debate-tone instruction, and token-probability responses, alongside a separate evaluation of final-answer quality.

Measuring disagreement. A change in reported agreement need not mean that a model’s position has changed. In our committee, members first form independent opinions and then respond to one another’s arguments. Each *debate turn* is a reply to a specific earlier contribution from another member, with recent exchanges from the same thread provided as context. Alongside its reply, the model reports how much it agrees with the contribution it is answering. A *stance instruction* sets the tone of these replies to friendly, neutral, or hostile (Section 3.2). The measurement problem is that this instruction may change the agreement label without producing a corresponding change in the text. For illustration, a model could label its reply as disagreement while writing, “I agree with your argument.” The label alone therefore cannot establish whether the model actually argues back, much less whether its stated position persists beyond the instruction. We address these questions through four layers of analysis: what disagreement the models report, what disagreement their replies express, whether dissent elicited under a hostile instruction persists after that instruction is removed, and, in a separate probe using open-weight models, how opposing arguments change the probabilities assigned to stance options (Figure 1):

- **LAYER A:** the agreement level the model reports about its own reply (what it *says* its stance is);
- **LAYER B:** whether the reply *text* actually pushes back (introduces a counter-argument, new evidence, or a refusal to endorse), judged by an external, condition-blind grader;
- **LAYER C:** does disagreement persist without the instruction that elicited it? For turns labeled as disagreement under the hostile instruction, we ask the same model to respond to the same argument again, keeping the debate context fixed. We repeat this with the instruction removed and with it retained, then compare how often the label reverts to agreement. This comparison separates the effect of removing the instruction from ordinary variation between repeated responses;
- **LAYER D:** how does an opposing argument change the model’s preference among possible stances? In a separate study with self-hosted open-weight models, we read the token probabilities assigned to seven stance options, ranging from strongly disagree to strongly agree. We compare these probabilities after an opposing argument with those after approximately length-matched filler text. This measures movement toward the opponent after adjusting for a uniform change in probability spread.

The main committee experiments focus on leading closed-weight models to study debate in high-capability systems, with Layers A, B, and C examining the same committee runs. For LAYER D, we conduct a separate study with self-hosted open-weight models and a proposition pool tailored to the stance probe. These models expose token probabilities unavailable from the main committee’s serving endpoints, allowing us to examine how opposing arguments change the option probabilities and movement along the stance scale. This analysis provides complementary evidence from a different model population, not a direct explanation of the closed-weight committee’s behavior. In the main experiments, changing the stance instruction substantially alters both reported agreement and the disagreement expressed in replies. Evidence that these changes persist after instruction removal is less conclusive, and separate evaluations provide limited evidence of improvements in final-answer quality. We therefore distinguish expressed disagreement, persistent positions, and better answers rather than treating them as interchangeable outcomes.

Contributions. We study three-model committees on open-ended opinion questions from GlobalOpinionQA [Durmus et al., 2023] and questions with verifiable answers from FRAMES [Krishna et al., 2024]. The complementary token-probability study uses a separate, authored set of policy propositions. Our committee implementation and all four evaluator layers are publicly available.¹ Our main contributions are:

1. **LAYER A: self-reported agreement.** We record how much each model reports agreeing with the argument it is answering, and compare these labels across friendly, neutral, and hostile debate conditions. In the three-round setting of our 750-run study, pooling replies across all three rounds, the share labeled as full agreement falls from 74.8% under friendly instructions to 24.4% under hostile instructions. A similar gap is already present after one round (Section 4.1). These comparisons establish how reported agreement varies with the instruction; whether disagreement persists without that instruction is tested separately in LAYER C.
2. **LAYER B: disagreement in the reply text.** We assess whether changes in reported agreement also appear in what the models write. A judge from outside the committee’s model families reads each reply together with the argument it answers, without seeing the structured self-reported label or the tone instruction. Its ratings reproduce the broad decline in agreement from friendly to hostile conditions in the main study. The effect therefore extends beyond the labels: the models also express more disagreement in their replies.
3. **LAYER C: a controlled test of persistence.** We test whether disagreement survives removal of the instruction that elicited it. For replies originally labeled as dissent under the hostile instruction, we ask the same model

¹<https://github.com/chenmoneygithub/llm-committee>

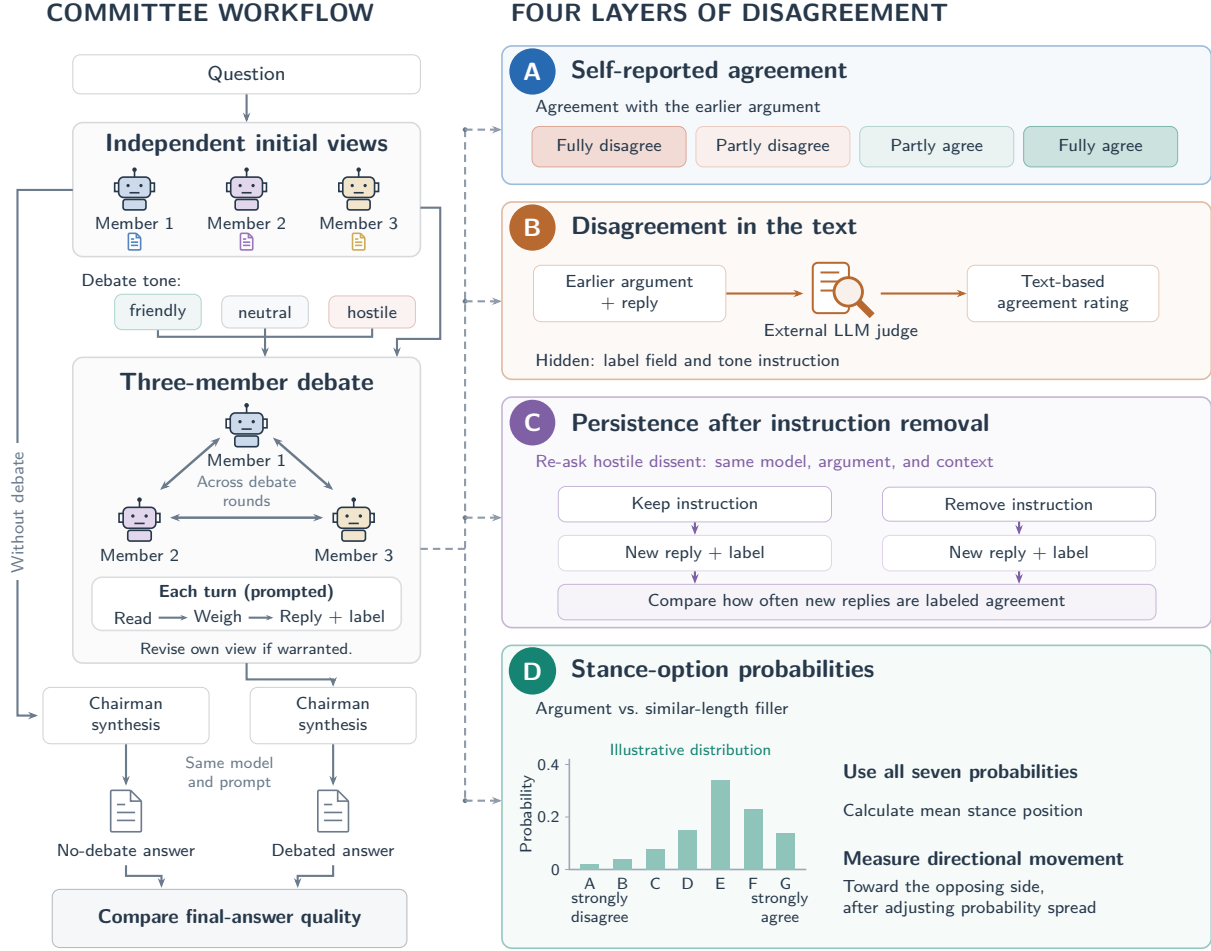


Figure 1: **Committee workflow and four-layer evaluation.** Members are prompted to read a peer’s argument, weigh it against their own view, and return a reply plus a self-reported agreement label. The triangle summarizes message flow across rounds; members refine their own views after responding to the received arguments. Layers A–C use the committee runs, while Layer D uses separate open-weight models and propositions. The probability distribution is illustrative.

to answer the same argument again, with the instruction either removed or retained. The retained-instruction control measures how often agreement changes simply because the model is asked again. The primary test provides suggestive rather than conclusive evidence of instruction dependence (Section 4).

- LAYER D: stance preferences from token probabilities.** We probe three open-weight models using seven stance options, ranging from strongly disagree to strongly agree. Rather than recording only the selected option, we read the token probabilities assigned to all seven option labels. We compare these readings after an opposing argument and after approximately length-matched filler text. After adjusting for probability spread, DeepSeek and Qwen show movement toward the opponent, while Llama’s directional response is unresolved. In DeepSeek and Qwen, strong, relevant arguments also produce larger directional shifts than forceful irrelevant arguments (Section 4.4).
- Downstream answer quality.** We compare final answers synthesized after debate with answers synthesized from independent initial views. A three-model jury evaluates each pair in both presentation orders to check for order effects. It returns no decisive preference in any of the 299 comparisons. However, tests with deliberately degraded answers show that the jury misses moderate quality differences, so these ties do not establish equal answer quality (Section 5).

The novelty is the *combination*, not “first to quantify committees.” We progressively separate reported, textual, persistent, and token-probability responses; use a judge from outside the committee’s model families, assessed using comparison judges; and benchmark that judge against two human annotators on a bounded sample (§4). We build on LMC [Zhao et al., 2024], ChatEval [Chan et al., 2023], Peacemaker [Yao et al., 2025], and Self-MoA [Li et al., 2025] (§2).

The takeaway, and what this paper does *not* claim. In plain terms, the findings are three. First, debate tone strongly changes how much agreement the committee *reports*, and the change is real in the reply text, not a labeling artifact (A, B). Second, the members engage with one another’s arguments, but we find much weaker evidence of *persistent* position change: part of the expressed dissent reverts once the instruction that elicited it is deleted, although the primary test is inconclusive (C). In the separate open-weight study, opposing arguments shift response probabilities toward the opponent in two models, without establishing lasting belief change (D). Third, we detect no quality gain from the debate, either on the open-ended pairs, where our jury detects tested severe damage but misses moderate differences, or on the verifiable control task. This is neither “debate works, done” (an over-read of LAYER B) nor “it’s all performance” (an over-read of the persistence probe, which is descriptive): on no-gold opinion items a position *change* cannot be scored as an *improvement* [Fanous et al., 2025], and our judge is validated only against a bounded two-annotator sample (Gate B, our judge-versus-human benchmark; not an equality or interchangeability test). The LAYER D instrument is a separate white-box probe built directly for the open-weight logprob grid.

2 Related Work

Debate and final-answer quality. Multi-agent debate serves two related but distinct purposes. In debate for oversight, a judge uses competing arguments to assess answers it may be unable to produce or verify unaided [Irving et al., 2018, Michael et al., 2023, Khan et al., 2024]. In debate for answer generation, models exchange and revise responses to improve their joint output [Du et al., 2023, Liang et al., 2023]. These studies cover more than exact-answer tasks: MAD also evaluates machine translation using automatic metrics and human ratings [Liang et al., 2023]. Here, *open-ended opinion questions* means questions without a unique correct answer, not simply tasks that require free-form text. Our main committee study uses such questions to examine disagreement separately from output quality.

Evidence that debate improves answers depends on the task, protocol, and baseline. Several evaluations find that debate does not reliably outperform a well-prompted single model or repeated independent sampling [Smit et al., 2023, Li et al., 2024a, Wang et al., 2024b, Zhang et al., 2025]. In an oversight setting, Kenton et al. [2024] find that debate outperforms direct question answering on extractive tasks where debaters see a source article that the judge cannot access; results on other tasks without this information asymmetry are mixed. That finding does not predict the outcome of our shared-context committee, whose final answer is synthesized rather than selected by a weaker judge.

The choice of baseline also determines what an improvement can establish. Mixture-of-Agents combines outputs through successive aggregation stages [Wang et al., 2024a], whereas Self-MoA shows that aggregating repeated outputs from a strong single model can outperform mixing models [Li et al., 2025]. Equal-budget comparisons can favor debate or aggregation at some budgets and self-consistency at others [Wunderlich et al., 2026]. Related studies examine limits imposed by shared model errors [Chen, 2026] and cases where a team underperforms its expert member [Pappu et al., 2026]; open-source council experiments also compare governance structures with self-consistency and prompt-diversity baselines [andybhall, 2025]. These results motivate separate comparisons for interaction, model composition, and computational cost. Equal sample counts, model-call counts, token counts, and dollar budgets are different controls. Our main comparison holds the committee roster fixed and varies peer interaction; it is not a matched-budget test against best-model aggregation.

Interaction schedules provide another source of variation. Repeated peer revision appears in early debate protocols [Du et al., 2023, Liang et al., 2023]; layered aggregation provides a related, but not identical, form of cross-model interaction [Wang et al., 2024a]. Our no-debate, single-round, and repeated-debate conditions compare these schedules within one committee implementation (see also Tran et al., 2025). Longer discussions need not be better: Kenton et al. [2024] find no significant benefit from additional turns in their tested protocols, while Becker et al. [2025] examine how discussion can drift away from the task. We therefore measure the consequences of interaction rather than treating more rounds or more agreement as evidence of improvement.

Disagreement, conformity, and persistence. Sycophancy studies show that models can align responses with a user’s stated views [Sharma et al., 2023, Perez et al., 2022]. Related work examines conformity under social influence [Zhu et al., 2024], the role of assertiveness in model-to-model debate [Xiong et al., 2023], and the propagation of sycophancy or loss of expressed disagreement across agents and rounds [Kasprova et al., 2026, Wynn et al., 2025]. Studies of user pushback and rebuttal timing likewise show why a changed answer need not reflect improved reason-

Study	Main measurement	Connection to this study
Du et al. [2023]; MAD [Liang et al., 2023]	Answer quality after exchanging and revising responses.	We separate answer quality from disagreement on opinion questions.
Smit et al. [2023]; Zhang et al. [2025]	Debate versus single-model and independent-sampling baselines.	Motivate testing, not assuming, the value of interaction.
Kenton et al. [2024]	Weaker judges using stronger debaters, with or without private evidence.	Oversight differs from joint-answer synthesis; their findings do not predict ours.
MoA; Self-MoA [Wang et al., 2024a, Li et al., 2025]	Quality of aggregated outputs, including same-model resampling.	Motivate separating interaction from extra sampling and aggregation.
LMC; PoLL [Zhao et al., 2024, Verga et al., 2024]	Agreement between model juries and human ratings.	Inform evaluator design; our human check concerns disagreement in replies.
ChatEval [Chan et al., 2023]	Human agreement when evaluators debate an external answer.	Their evaluators debate; ours rates disagreement in the debate under study.
Peacemaker [Yao et al., 2025]	Disagreement collapse, correct-position abandonment, and sycophancy.	Our persistence test concerns instruction dependence, not correctness of a change.
Hao et al. [2026]	Answer changes after self-reflection, peer stances, or peer reasoning.	Both control repeated answering; we vary tone-instruction retention.
Keramati et al. [2026]	Reasoning-token probabilities, judge ratings, and accuracy.	We measure adjusted movement along a fixed stance scale.
This study	Reported and textual agreement, persistence, and stance-option probabilities.	Answer quality is a separate outcome; the token probe uses other models and propositions.

Table 1: Selected studies, their primary measurements, and their relationship to our four-layer analysis. Opinion questions without a unique correct answer are distinct from free-form generation tasks such as translation.

ing [Guo et al., 2026, Fanous et al., 2025]. These findings motivate our comparison of friendly, neutral, and hostile instructions, but the interventions are not interchangeable: a peer’s answer, a dissenting participant, and a standing tone instruction provide different kinds of input.

Peacemaker or Troublemaker [Yao et al., 2025] connects debate dynamics to correctness and sycophancy under different personas. Its Negative Agreement Rate measures abandonment of a correct position during disagreement. Our persistence test instead asks whether dissent remains after the instruction that elicited it is removed. On opinion questions, remaining consistent is not necessarily correct, and changing position is not necessarily an error. The two tests therefore address different questions; persistence is not a replacement for a correctness-based measure.

A closely related measurement problem is that an answer can change simply because the model is asked again. Hao et al. [2026] compare self-reflection, exposure to peers’ stances alone, and exposure to their full reasoning to distinguish spontaneous answer changes from peer influence. Our LAYER C comparison varies a different input: we re-ask the same model with the debate context fixed and the tone instruction either removed or retained, then examine both the agreement label and the reply text. The retained-instruction response controls for variation between repeated answers.

Work on instruction stability and repeated questioning provides further reasons to test whether a stated position persists [Li et al., 2024b, Xie et al., 2023]. Faithfulness studies show that generated explanations need not reveal what influenced a response [Turpin et al., 2023, Lanham et al., 2023]. We draw a narrower distinction: the model’s agreement label, disagreement expressed in its reply, and persistence after instruction removal are separately measured outcomes. None alone establishes an underlying belief or the reasoning process that produced the answer.

Evaluating replies and final answers. A model committee can also act as an evaluator rather than as the system being evaluated. Language Model Council compares aggregated model judgments with human preferences on subjective tasks [Zhao et al., 2024]; PoLL combines judges from different model families [Verga et al., 2024]. ChatEval uses discussion among evaluators to assess an external answer [Chan et al., 2023], while DEBATE uses a devil’s-advocate evaluator to challenge an initial assessment [Kim et al., 2024]. These methods inform our use of model judges, but evaluating a completed answer and measuring disagreement within the discussion are different tasks.

We use one instrument to rate disagreement in replies and a separate jury to compare final answers. For the former, a judge from outside the committee’s model families sees the reply and the contribution it answers, but not the self-reported agreement label or tone instruction. We compare its ratings with other model judges and with hu-

man annotations. This is a task-specific comparison, not evidence that model judges can generally replace people; JUDGE-BENCH documents substantial variation across evaluation tasks [Bavaresco et al., 2024]. For final-answer comparisons, presenting each pair in both orders follows established work on position bias [Wang et al., 2023, Zheng et al., 2023]. We additionally test the jury on deliberately degraded answers to determine which quality differences it can detect. Order consistency alone cannot show that a judge is sensitive enough to support an equivalence claim.

Token probabilities and stance options. Keramati et al. [2026] study the relationship between log probabilities of generated reasoning tokens, external judgments of that reasoning, and final-answer accuracy. Our LAYER D probe instead reads the probabilities assigned to a fixed set of seven stance options, from strongly disagree to strongly agree. We measure adjusted movement toward the opposing side along that stance scale. Comparisons with approximately length-matched filler and with forceful but irrelevant arguments help distinguish responses to an argument’s content from other effects of adding text. These are contextual response probabilities, not calibrated confidence or direct measurements of beliefs. The probe uses a separate open-weight model population and proposition pool, so it complements, rather than explains, the closed-weight committee results (Section 4.4).

Together, these literatures motivate our separation of reported agreement, disagreement in text, persistence after instruction removal, and stance-option probabilities (Table 1). We evaluate final-answer quality separately because neither expressed disagreement nor a change in stance establishes that the committee produced a better answer.

3 Experimental Setup

3.1 Tasks and committee workflow

Our main experiments use opinion questions from GlobalOpinionQA [Durmus et al., 2023]. These questions have no unique correct answer. We therefore evaluate how models disagree and what their final answers contain, without treating the survey majority as ground truth. To also evaluate correctness, we use FRAMES [Krishna et al., 2024], a multi-hop factual question dataset, and GPQA [Rein et al., 2023], an expert-level question dataset.

For each question, we run a committee of three LLM participants, which we call *members*. The main committee consists of Claude Opus 4.8, GPT-5.5, and Claude Sonnet 4.6, denoted *tri3* in the artifacts. Each member first answers the question independently, without seeing the others’ responses. The members then discuss these initial answers over a fixed number of rounds. Members have the same web-search and page-fetch tools with and without debate on all three tasks.

On each *debate turn*, a member replies to one specific contribution from a peer. Its input includes that contribution, its own current view, and recent exchanges from the same discussion thread. The output contains both a written reply and a label indicating how much the member agrees with the contribution it is answering. During each round, members respond to their assigned peer contributions and produce refined versions of their own answers.

For example, on the university-education question below, GPT-5.5 rejects the claim that education matters more for boys. Claude Opus 4.8 agrees with that answer but challenges one of GPT’s reasons, labeling its reply *partially agreed*. The label concerns agreement with the peer’s message, not agreement with the question’s claim. Appendix A.2 shows this actual exchange and the text judge’s rating.

At the end, Claude Opus 4.8 acts as the *chairman*, synthesizing the members’ final views into one answer. The no-debate baseline uses the same chairman but skips the exchange of replies, sending the independent initial answers directly to the chairman. It therefore retains multiple members and synthesis; it is not a solo-agent baseline. Appendix A gives the routing and implementation details. The following experiments vary the instructions and number of rounds within this workflow. The separate protocol needed to read token probabilities is described in Section 3.5.

3.2 Experimental conditions

We vary both how members are instructed to engage with one another and how long they discuss. The first manipulation is a standing *tone instruction*, applied to debate-turn prompts:

- **Friendly:** seek common ground and concede when the other contribution has merit.
- **Neutral:** use the base debate prompt without an additional tone instruction.
- **Hostile:** stress-test each position and avoid agreement without a substantive reason.

Initial drafting and chairman synthesis use neutral instructions in every condition. Thus the manipulation concerns engagement with peers, not the dataset answer options or the chairman’s task. The exact tone additions appear in Appendix A.1.

Main study. We select a fixed set of 50 questions from GlobalOpinionQA, a dataset of public-opinion survey questions [Durmus et al., 2023]. We retain questions that ask for opinions rather than reports of respondents’ personal experiences, habits, or identities. For example, we include a question asking whether respondents agree with the statement “A university education is more important for a boy than for a girl,” but exclude the personal-history question “In the last five years have you traveled to another country other than Canada or Mexico?” Three LLMs assess the candidate questions; a question is eligible if at least two classify it as an opinion question (Appendix A.2). Each selected question is evaluated under friendly, neutral, and hostile instructions with one debate round, and under those same three instructions with three debate rounds. The question text, answer options, and committee models remain identical across these six runs; initial answers are generated anew for each run. For each question, we also perform three no-debate runs, in which the members’ independent initial answers go directly to the chairman without any exchange of replies or tone instruction.

Reference-answer tasks. We separately evaluate final-answer accuracy on FRAMES and GPQA. For each dataset, we use one fixed set of 50 questions. Each question is evaluated without debate, with one neutral debate round, and with three neutral debate rounds.

3.3 Reported and textual agreement: Layers A and B

We first ask whether the member’s agreement label matches what its reply actually expresses. Both readings concern agreement with the contribution being answered, not the member’s position on the underlying question: a member can dispute a peer’s reasoning without changing its own answer.

Self-reported agreement (A). Each reply includes an `agreed_level`: fully agreed, partially agreed, partially disagreed, or fully disagreed. We compare these label distributions across conditions, focusing on the share of replies labeled fully agreed. Only explicitly emitted labels enter the analysis. If the output schema supplies a default because the model omitted the field, we flag and exclude that label. The accompanying `opinion_shift` field reports change in the member’s own position, from 0 (no change) to 4 (complete reversal); it is not an independent measurement.

Agreement expressed in text (B). To assess the reply separately from its label, a model judge reads the contribution and reply without seeing the member identity, tone condition, or structured self-report. Self-descriptions embedded in the reply itself are not removed (Appendix A.2). It rates agreement on the same four-level scale. The judge is Gemini 3.5 Flash, from a different model family than any main-committee member: Claude Opus 4.8 and Claude Sonnet 4.6 are Anthropic models, and GPT-5.5 is an OpenAI model. The judge also classifies the reply as a challenge, new argument, concession, refinement, or restatement. This describes the conversational action, not just agreement: a new argument, for example, can support the peer’s conclusion. The rubric is summarized in Appendix B.

Checking the model judge. To assess Gemini 3.5 Flash’s agreement ratings, we ask Claude Opus 4.8 and GPT-5.5 to independently rate the same texts. We sample 36 replies from earlier mixed-model committee runs, outside the main-study outcome sample. The sample contains equal numbers of replies self-labeled as agreement and disagreement, so both are represented. These self-reported labels are used only for sampling. For each item, all three judges see the same reply and the peer message it answers, without the member identity, tone instruction, or structured self-reported label. Each judge rates agreement on the same four-level scale. We compare Gemini’s ratings separately with Opus’s and GPT’s using Gwet’s AC1, a chance-adjusted agreement coefficient with 1 indicating identical ratings [Gwet, 2008]. We also inspect whether Gemini uses different labels rather than almost always choosing one. This checks consistency across model judges, not accuracy against a gold label. The calculation and validation results are in Appendix B.

Comparing model and human ratings. Each human annotator sees the original question and a two-message exchange between committee members: one member’s message and another member’s reply. The annotator labels whether the reply fully agrees, partially agrees, partially disagrees, or fully disagrees with the preceding message. The member’s structured self-reported label is hidden, and the two annotators work independently.

We then ask whether Gemini’s ratings match a human reader’s ratings as closely as another human reader’s do. One annotator’s labels serve as the common reference. We calculate how closely Gemini matches these labels, and separately how closely the second annotator matches them, using the same replies for both comparisons. The reported

difference subtracts the second annotator’s score from Gemini’s: a negative value means Gemini matches the reference less closely.

3.4 Dependence on the tone instruction: Layer C

Layers A and B compare reported and textual agreement across debates conducted under different tone instructions. Layer C asks a narrower question: for a case where a model originally expressed disagreement, does removing the hostile instruction make a newly generated reply more likely to agree? We test this by changing the instruction while holding the question, the message being answered, and the model’s preceding input context fixed. This follows work on repeated questioning and response stability [Xie et al., 2023, Li et al., 2024b].

Our primary analysis uses first-round replies labeled partially or fully disagreed under the hostile instruction. At this stage, the member’s own initial answer and the message it responded to are available in the saved debate record. For each selected case, we ask the same model to generate a new reply twice: once with the hostile instruction retained and once with it removed. Both calls present the same selected exchange and the member’s initial answer. The original reply is used to select the case but is not shown in either call.

We count a new label of partial or full agreement as a change from disagreement to agreement. The retained-instruction call measures how often this happens simply because the model generates another response. For each question, we calculate the fraction of selected replies changing to agreement in each condition, subtract the retained-instruction rate from the removed-instruction rate, and average these differences across questions. A positive difference means that instruction removal produces additional changes to agreement. As a supplementary control, we repeat neutral-condition cases twice with identical prompts and subtract their between-draw difference on shared questions.

Later-round cases are secondary because the member’s updated own-view input was not saved; we use its latest earlier message as a proxy. Their inputs also contain messages generated during earlier hostile rounds, which remain unchanged when the current instruction is removed. The probe therefore tests immediate dependence on the explicit instruction, not lasting belief or the model’s underlying reasoning [Lanham et al., 2023].

3.5 Stance-option probabilities: Layer D

Layer D quantifies responses to an opposing argument through the probabilities of fixed agree/disagree options, rather than one generated answer [Kadavath et al., 2022, Burns et al., 2023, Santurkar et al., 2023, Xiong et al., 2024].

Models and statements. We use locally served Llama-3.3-70B-Instruct, Qwen3-235B-A22B, and DeepSeek-V3, with reasoning disabled, because the main committee’s endpoints do not expose the required probabilities. We author 36 candidate statements, such as “Space exploration is a worthwhile use of public funds.” These share one agree/disagree scale; GlobalOpinionQA would require item-specific option mappings. Statements are selected separately for each model before any discussion (Appendices D and D.3).

One exchange and its control. A model first writes its view on a statement. With that answer in context, we ask it to rate the *original statement* with one letter A–G: A means strongly disagree, G strongly agree, and B–F are intermediate positions. We read all seven letters’ log probabilities at the first answer-token position.

A second model reads the statement and initial answer and is asked to argue the opposite position, as determined by the initial A–G reading. We request either an evidence-rich counterargument (*strong*) or a vague, unsupported one (*weak*). To test whether addressing the topic matters, a third condition instead requests a forceful argument about an unrelated statement (*irrelevant*). Each of these three requests is tested with friendly and hostile delivery. For strong and weak arguments, both tones request the same opposing position; “strong” and “weak” describe the prompts, not independently verified argument quality.

We then give the first model two alternative inputs, each ending with the same A–G rating request:

- **Argument:** the original statement, its initial answer, and the second model’s message.
- **Control:** the same statement and initial answer, but with a procedural message in place of the argument.

The control repeats a procedural sentence to approximately match the argument’s token length, without discussing the statement. This provides a baseline for extending the conversation without taking a position on the topic. We read the A–G probabilities separately for each input; the model does not first write a response to the added message. Each new trial generates a fresh initial answer. Every model faces both other models (prompts in Appendix D.1).

From letter probabilities to a stance score. Index A–G by $k = 1, \dots, 7$, and let ℓ_k be a letter’s recorded natural-log probability. Define

$$p_T(k) = \frac{\exp(\ell_k/T)}{\sum_{j=1}^7 \exp(\ell_j/T)}, \quad \mu(p_T) = \sum_{k=1}^7 k p_T(k).$$

Here μ is the average position on the equally spaced stance scale. T is the *read temperature*, used to rescale saved probabilities, not to generate messages. We use $T = 2$ to reduce concentration on a single letter; for example, odds of 100:1 become 10:1. Effects can depend on this choice, so Appendix D.3 also reports $T = 1$ and $T = 4$.

Change in mean stance rating. Let p_T^{argument} and p_T^{control} be the distributions from the argument and filler inputs. The direct difference in their mean ratings is

$$\begin{aligned} \Delta &= s[\mu(p_T^{\text{argument}}) - \mu(p_T^{\text{control}})], \\ s &= \begin{cases} +1, & \text{assigned side: agreement,} \\ -1, & \text{assigned side: disagreement.} \end{cases} \end{aligned} \quad (1)$$

The assigned side is determined from the initial reading on the original statement; unrelated messages retain the same reference direction. The sign makes a change toward that assigned side positive. It does not measure whether the model agrees with the opponent’s reasoning.

Comparison at matched entropy. We also compare mean ratings after adjusting the control to have the same entropy as the argument distribution. This holds probability concentration, as measured by entropy, constant while preserving the control’s option ranking. For the seven options, define entropy H and the adjusted reference q as

$$\begin{aligned} H(p) &= -\sum_{k=1}^7 p(k) \log_2 p(k), \\ q(k) &= \frac{[p_T^{\text{control}}(k)]^{1/h}}{\sum_{j=1}^7 [p_T^{\text{control}}(j)]^{1/h}}, \quad H(q) = H(p_T^{\text{argument}}). \end{aligned}$$

We choose $h > 0$ to satisfy the entropy equality. The argument probabilities set the target entropy; the control probabilities supply the ranking that q preserves. Equal entropy does not require equal option probabilities. The reference is calculated, not another model response. The entropy-matched difference is

$$D = s[\mu(p_T^{\text{argument}}) - \mu(q)]. \quad (2)$$

Both scores use stance-scale points. Δ includes the entire argument–filler mean difference; D subtracts the part produced by this particular rescaling of the control. Neither score establishes a change of side or lasting belief revision. Appendix D.2 gives a worked example.

Analysis. We report both scores on the same statement samples. For each statement, we average its strong and weak counterargument measurements. A separate content comparison subtracts the score for an unrelated message from that for a strong counterargument, matching model, opponent, statement, and tone. We then average these statement-level values, weighting each statement equally. Table 13 gives the model-specific samples; selection and statistical procedures are in Appendices D and D.3. The original tests concern D ; direct differences are added descriptively from the saved probabilities.

3.6 Final-answer quality

Disagreement during discussion need not improve the answer produced at the end. We use pairwise quality judgments for opinion questions and reference-answer correctness for factual questions.

Comparing open-ended answers. We compare each debated synthesis with the same committee’s no-debate synthesis for that question. A jury of Claude Opus 4.8, GPT-5.5, and Gemini 3.5 Flash scores each answer from 1 to 5 on coverage, treatment of opposing views, commitment to a position, and soundness. Each judge prefers the higher summed score only when the difference exceeds two points; otherwise it returns a tie.

Every answer pair is shown in both orders to check whether a judge prefers an answer or merely its position [Wang et al., 2023, Zheng et al., 2023]. For example, preferring the debated answer both when it appears first and when it appears second is consistent. Preferring whichever answer appears first is not. A judge’s verdict is retained only when

Rounds	Tone	Replies	Fully agreed	Partially agreed	Partially disagreed	Fully disagreed
One	Friendly	300	68.0	30.0	2.0	0.0
	Neutral	300	49.0	48.0	3.0	0.0
	Hostile	300	23.0	69.7	7.3	0.0
Three	Friendly	824	74.8	23.2	2.1	0.0
	Neutral	860	57.2	40.0	2.8	0.0
	Hostile	837	24.4	67.1	8.5	0.0

Table 2: **Layer A: members’ self-reported agreement with the peer message.** Each percentage divides the label count by the eligible replies in that row; rows sum to 100% up to rounding. Three-round rows pool replies from all three rounds. We exclude automatically supplied labels: 7 friendly, 2 neutral, and 10 hostile replies in the three-round arm; none in the one-round arm.

both presentations favor the same answer or both return a tie; otherwise its vote becomes a tie. The jury returns the strict majority verdict, or a tie if there is none. All main-study answer pairs remain in the analysis. Unlike the reply judge, this jury includes families represented in the committee. The matched answer rosters reduce a composition asymmetry but do not guarantee unbiased judgments [Wataoka et al., 2024].

An always-tie judge would pass an order check without distinguishing quality. We therefore also compare original syntheses with versions deliberately damaged at severe, moderate, and subtle levels, ranging from replacing the answer to deleting one supporting sentence. For this separate check, we retain pairs on which all three judges are consistent across presentation orders. Detection is the fraction of retained pairs in each damage level for which the jury prefers the original. This checks whether ties could reflect missed quality differences (Appendix E.1).

Scoring answers with a known reference. On FRAMES and GPQA, Claude Opus 4.8 reads the question, reference, and final synthesis. It marks an answer correct if it contains the reference answer without contradiction, incorrect if it contradicts it, and not attempted if it does neither. Accuracy is the correct fraction of completed runs; API failures are excluded. This grader shares a model with the committee. The separate mathematical-task control is described in Appendix E.2.

4 Results

Reading the agreement tables. Tables 2 and 3 report percentages of individual replies in each agreement category, not final committee answers. One-round rows contain that round’s replies; three-round rows pool all three rounds, not just the last. Differences between percentages are in percentage points (pp).

4.1 Layer A: tone shifts full agreement toward qualified agreement

Friendly instructions produce the most full-agreement labels and hostile instructions the fewest, with neutral instructions in between (Table 2). In three-round debates, the friendly and hostile rates are 74.8% and 24.4%, a 50.4pp difference (95% CI [42.3, 58.2]). With one round, the difference is already 45.0pp ([34.7, 54.7]). Intervals resample questions rather than individual replies (Appendix A.2).

The full distribution clarifies what replaces full agreement. Under hostile instructions, most replies are still labeled *partially agreed*: 69.7% with one round and 67.1% with three. Partial disagreement accounts for 7.3% and 8.5%, respectively; no reply explicitly reports full disagreement. Tone therefore changes the strength of reported agreement much more than it produces self-reported rejection. These are comparisons between fixed-tone runs, not successive decreases at each turn.

4.2 Layer B: the text contains more disagreement than members report

Gemini 3.5 Flash independently rates the peer message and reply without seeing the tone instruction or structured self-label. Its four-level ratings show the same friendly–neutral–hostile ordering in both schedules (Table 3). Full disagreement remains rare, but partial disagreement is more common than the members’ labels suggest. For example, among the same 300 hostile one-round replies, Gemini rates 28.3% as partially disagreed, compared with 7.3% in the self-reports.

Rounds	Tone	Replies	Fully agreed	Partially agreed	Partially disagreed	Fully disagreed
One	Friendly	300	65.7	27.3	7.0	0.0
	Neutral	300	41.7	44.7	13.7	0.0
	Hostile	300	23.3	48.3	28.3	0.0
Three	Friendly	831	75.2	20.1	4.7	0.0
	Neutral	862	59.6	31.9	8.5	0.0
	Hostile	847	27.6	45.8	26.1	0.5

Table 3: **Layer B: Gemini 3.5 Flash’s agreement ratings of the reply text.** Percentages use all judged replies in each row, including replies without an explicit self-label. The judge sees the peer message and reply, not the tone condition or structured self-label. Three-round rows pool all three rounds. Each condition uses the same 50 questions; one friendly three-round run failed. These are percentages of replies, not of questions.

Across the 3421 replies with both an explicit self-label and a judge rating, the four-level labels match in 78.5% of cases. The judge assigns a more disagreeing category in 15.3% and a less disagreeing category in 6.2%. These percentages divide the respective counts by the same paired-reply total. Thus the tone pattern appears in the reply text. Relative to this judge’s reading, self-labels more often understate than overstate disagreement. This is agreement with a peer message, not a change in the member’s answer to the original question.

Human comparison remains a separate validation check, not gold labels for these main-study replies (Table 7; Appendix B.2).

4.3 Layer C: re-asking with and without the instruction

A/B compare different tone conditions. C instead takes cases where a model originally reported disagreement under hostile instructions and gives the same model the same question and discussion context twice: once with the hostile instruction removed and once with it retained. We count new replies labeled partially or fully agreed with the peer message.

The sample contains 39 first-round reply contexts from 19 questions. For each question, we divide the number of new agreement labels by the number of tested contexts, separately for removal and retention. Averaging the removal-minus-retention differences across questions gives +11.8pp. 5 questions have a higher agreement fraction after removal; 14 have the same fraction in both conditions. This is an observed difference in newly generated reply labels, not a measure of lasting endorsement of the underlying position. The question-level test, repeat-response control, and approximate later-round comparison are in Appendix C.

4.4 Layer D: arguments change the mean stance rating

Layer D measures the expected rating of the original statement, not whether a model thinks the opponent is correct. We encode A–G as 1–7 and average using their probabilities at $T = 2$. The direct score Δ compares this mean after an argument with its mean after filler. The entropy-matched score D instead uses the calculated reference q . Both signs are oriented toward the side assigned to the opponent: a lower mean counts positively when disagreement was requested, and a higher mean counts positively when agreement was requested (Section 3.5).

Direct and entropy-matched differences. On the same model-specific statement samples, the direct mean differences are +1.47 scale points for DeepSeek, +0.75 for Qwen, and +0.20 for Llama (Table 4). After entropy matching, these become +1.19, +0.43, and +0.06. Thus the reference adjustment reduces all three estimates; the two columns answer different questions and should not be treated as interchangeable. The multiple-comparison-adjusted tests support a positive entropy-matched difference for DeepSeek and Qwen, but do not resolve Llama’s smaller adjusted estimate. The direct differences are descriptive analyses of the same saved probabilities, with uncertainty reported separately (Appendix D.3).

Does addressing the statement matter? We compare evidence-rich counterargument requests with forceful arguments about an unrelated statement, matching the measured model, opponent, statement, and tone. The second panel subtracts the unrelated-message score from the counterargument score, separately for Δ and D . Both contrasts are positive in all three models at $T = 2$. For the entropy-matched contrast, DeepSeek and Qwen retain the effect across the checked temperatures; Llama’s small contrast does not (Appendix D.3). The labels describe argument requests, not independently verified quality.

Model	Statements	Direct difference Δ	Entropy-matched difference D
Change in mean stance rating: strong and weak counterarguments			
DeepSeek-V3	15	+1.47	+1.19
Qwen3-235B	15	+0.75	+0.43
Llama-3.3-70B	8	+0.20	+0.06
Relevant versus unrelated message: strong-minus-unrelated difference			
DeepSeek-V3	17	+1.16	+1.14
Qwen3-235B	15	+0.91	+0.54
Llama-3.3-70B	19	+0.20	+0.07

Table 4: **Layer D: changes in the probability-weighted A–G rating.** Values are signed scale points at $T = 2$, positive toward the assigned side. Δ compares argument and filler means; D uses the entropy-matched reference q . The lower panel subtracts unrelated-message scores from strong-counterargument scores. Columns use the same observations and weight statements equally. Intervals and tests: Appendix D.3.

Reference-answer tasks: accuracy (%), neutral tone			
Task	No debate	One round	Three rounds
FRAMES ($n = 50$ per arm)	94.0	92.0	92.0
GPQA ($n = 38$ per arm)	97.4	97.4	94.7
AIME 2026 + HMMT Feb. 2025	100.0 (47/47)	Not run	100.0 (46/46)
Open-ended questions: pairwise quality judgments			
Task	Baseline	With debate	Jury ties
	Independent drafts + synthesis	One or three rounds + synthesis	
GlobalOpinionQA ($n = 50$)			299/299

Table 5: **The committee’s final answers with and without debate.** Accuracy is correct/completed graded answers; failed calls are excluded. All reference-task arms end in synthesis. Math parentheses give correct/graded counts: 45 problems overlap; 13 slow problems were uncollected in both arms (tools disabled). GlobalOpinionQA pools both schedules and all tones; ties are inconclusive because the jury misses tested moderate damage. Near-ceiling reference-task scores also do not establish equivalence.

These are changes in the probability-weighted response to a statement. A positive score need not mean crossing from agreement to disagreement, endorsing the opponent’s reasoning, or retaining a new position outside this context. Each estimate averages repetitions for a statement first, then gives each statement equal weight.

4.5 Final answers: does debate improve quality?

More disagreement need not produce a better answer. We therefore compare the chairman’s synthesis after debate with its synthesis of the same committee’s independent initial drafts, holding the question and model roster fixed. This isolates the addition of debate to a committee, not the value of a committee over a single model. Table 5 summarizes the committee’s final-answer comparisons on open-ended, factual, and mathematical tasks.

The open-ended jury detects no advantage, but misses moderate changes. Across both debate schedules and all tones, the jury returns a tie on all 299 answer pairs. A tie means that the scoring and voting procedure produces no decisive preference: scores within the rubric’s tolerance, order-inconsistent votes, and lack of a majority can all yield a tie (Section 3.6). On 214 pairs, every judge’s verdict is unchanged when the answer order is reversed, so the all-tie result is not solely caused by converting unstable votes to ties.

To interpret this result, we test the same jury on original answers paired with deliberately damaged versions. Among pairs passing the order check, it prefers the original in all 16/16 severe cases, but in none of the 12 moderate or 15 subtle cases. Severe changes replace the answer with one from another question or heavily truncate it; moderate changes remove a consideration or the stated position. The jury therefore detects the tested severe damage but misses meaningful smaller changes. The main result is *no detected quality advantage under this evaluator*, not equal answer quality (Appendix E.1).

High factual accuracy leaves little room to distinguish the schedules. On FRAMES and GPQA, accuracy is the fraction of completed final answers judged correct against the reference; API failures are excluded. The FRAMES and GPQA rows compare no debate, one round, and three rounds under neutral instructions. FRAMES accuracy differs by one correct answer per 50 questions between no debate and either debated schedule. GPQA is also near ceiling. These tasks provide no clear evidence of an accuracy gain, but the high starting accuracy and limited sample do not establish equivalence. On the collected AIME 2026 and HMMT February 2025 problems, both schedules answer every graded problem correctly; slow, uncollected problems are not represented. The table reports the denominators rather than treating this subset as complete benchmark coverage. Mathematical-task collection details are in Appendix E.2.

A separate two-agent web-research experiment is reported in Appendix E.3; its different roster and protocol are not pooled with these committee results.

5 Limitations

Scope and sample coverage. The main study uses one committee roster and discussion structure on GlobalOpinionQA. Its no-debate baseline retains all members and synthesis, so it does not compare debate with an equal-budget solo model. The probability probe uses different models and a small, response-selected proposition pool, not the internals of the main committee. Near-ceiling factual accuracy limits detectable improvements, and the math results exclude uncollected slow problems. High GPQA accuracy alone likewise does not establish contamination.

Model judgments and human comparison. Blinding hides metadata, not self-descriptions embedded in reply text. Cross-model checks reduce particular judging risks, not all systematic error. The text judge’s human comparison covers 107 replies from 43 development-sample questions, over-representing hostile turns. An author and an adult family member supplied the labels. The comparison does not establish that Gemini can replace a human annotator (Table 7; Appendix B.2). The main study’s quality jury and reference grader share model families with its answer generators. The jury misses tested moderate and subtle damage, so its ties are not evidence of equal quality. Final-answer grading remains AI-only.

Interpreting response changes. C measures immediate re-asking in reconstructed contexts, not cross-session persistence; its primary sample contains only 19 questions. Later-round contexts require an approximate previous view (Appendix C). D measures prompt-conditioned answer probabilities on an equally spaced stance scale. Its directional adjustment removes changes explained by one rescaling model, not every alternative to persuasion. Endpoint concentration can limit sensitivity, and directional magnitudes depend on the read temperature and stance encoding (Appendix D). Neither probe establishes lasting belief change. On opinion questions, even an observed position change is not itself a correctness improvement [Fanous et al., 2025].

6 Conclusion

The measurements answer different questions about multi-agent debate. Friendly, neutral, and hostile instructions change both members’ agreement labels and the interaction visible in their replies. This is evidence of a change in expressed agreement, not by itself a change in members’ answers to the question. Removing the hostile instruction produces more agreement in our sample, but the primary test remains inconclusive.

In the separate open-weight study, opposing arguments shift DeepSeek’s and Qwen’s mean stance ratings toward the opposing side, even after adjustment for probability spread; Llama’s adjusted result remains unresolved. In DeepSeek and Qwen, relevant arguments produce larger adjusted changes than unrelated ones, and these directions hold across the tested read temperatures. These are changes in which answers a model is likely to give under a particular prompt, not evidence of lasting belief change.

For final answers, the open-ended jury detects no advantage over synthesis of independent drafts, but its limited sensitivity leaves meaningful quality differences unresolved. Near-ceiling factual controls are similarly inconclusive about equivalence.

The contribution is a measurement framework that keeps these conclusions distinct. LLM debate readily changes what agents say, but we find much weaker evidence that it changes what they persistently endorse or improves the quality of the final answer.

Ethics Statement

This study manipulates the *stance* with which LLM committee members debate opinion statements drawn from GlobalOpinionQA, a public benchmark of cross-national survey questions. The manipulation targets debate style (cooperative versus adversarial), not the content of any opinion; the paper reports measurement results about model behavior and makes no normative claim about the survey propositions themselves. No human subjects were deceived: the only human data are annotation labels produced by two annotators who knowingly labeled machine-generated debate turns through a web application (Appendix B.2), with the models’ structured self-reported labels withheld by the interface. The annotators are one author and one adult family member of that author who is not an author; both participated as uncompensated volunteers and followed the written annotation codebook. The author-annotator is necessarily not blind to the study’s hypotheses; the family-member annotator was not briefed on them. Accordingly, Gate B uses these annotations only to compare the automatic judge’s agreement with humans against the two humans’ agreement with each other, never as ground truth, and this bounded, annotator-dependent scope is stated in Limitations. All debate transcripts, syntheses, and judge verdicts analyzed here are LLM-generated, and the paper discloses throughout which judgments are AI judgments; the web-research extension’s final-answer grading is AI-only and is scoped as such. Benchmark plaintext for the LiveBrowseComp extension is withheld from release under the benchmark’s canary policy.

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A Committee instructions and a worked example

Implementation details. Each member answers at most two incoming threads per round in one call, returning a reply to each and one updated answer. Search and page-fetch tools are used during initial drafting; peer exchanges are text-only. Member and chairman temperature is 1.0. The chairman template is identical with and without debate, including its reference to prior discussion. Software-defaulted agreement labels are excluded from Layer A, but their reply texts remain eligible for Layer B.

A.1 Tone additions

Only the debate instruction receives the following additions. Neutral adds nothing; drafting and synthesis do not receive either addition.

Friendly (verbatim).

DEBATE STANCE — COOPERATIVE: Approach this debate looking for common ground. Where another member makes a reasonable point, acknowledge it and build on it. Prefer synthesis and consensus; concede points readily when the other side has merit. Only maintain disagreement when you have a strong, specific reason.

Hostile (verbatim).

DEBATE STANCE — ADVERSARIAL: Your job is to stress-test every position, including your own. Actively look for weaknesses, hidden assumptions, missing evidence, and counterexamples in other members’ arguments. Do NOT agree merely to be agreeable or to reach consensus — agreement is only acceptable when

you are genuinely convinced after trying hard to refute. Steelman the opposing view, then attack it. If you still hold a different position, say so plainly and defend it. Productive disagreement is the goal; premature consensus is a failure.

A.2 Agreement with a reply versus agreement with a statement

This first-round exchange is from a hostile, three-round GlobalOpinionQA run. Excerpts retain the original wording; [...] marks omissions.

Question.

Do you agree or disagree with the following statement? A university education is more important for a boy than for a girl.

GPT-5.5's initial answer.

I think a university education is not inherently more important for a boy than for a girl. Both have equal rights to education, equal capacity to benefit from it, and equal claims to personal and professional development. [...] If anything, expanding girls' access to university may be especially socially important where girls have historically been excluded.

Claude Opus 4.8's reply.

I agree with your conclusion (completely disagree) [...] I'll push on one point, though. [...] The most robust basis for 'completely disagree' is the principled one you also state: equal moral worth and equal claims to intellectual development, which does not depend on any contingent return comparison. I mark this `partially_agreed` only because I want the principled claim to be load-bearing rather than the empirical asymmetry.

Both Opus's self-label and Gemini's text rating are *partially agreed*: the reply accepts GPT's conclusion but disputes a reason. Both models still reject the original statement. Gemini sees the full messages without model identities, tone, or the structured label. Phrases embedded in the reply, such as "I mark this `partially_agreed`," remain visible; the rubric tells the judge to assess substance, not meta-commentary.

Question screening. Claude Opus 4.8, GPT-5.5, and Gemini 3.5 Flash each assess whether the question and answer options express a debatable opinion rather than personal circumstances. Eligibility requires two `debatable_opinion` votes; the judges' confidence scores do not affect selection.

Uncertainty for agreement rates. The Layer A intervals resample questions independently within each tone and keep all replies belonging to each sampled question [Efron, 1979]. Each replicate recomputes the pooled reply percentages and their friendly–hostile difference, rather than averaging question-specific percentages.

B Reply judge rubric and validation

B.1 Rating rubric

Gemini 3.5 Flash judges the reply against the peer message's central claim, ignoring role or tone meta-commentary, length, and confident wording. The four anchors are:

Fully disagreed: rejects the central claim and argues for an opposing position.

Partially disagreed: contests part of the claim or accepts it only with major reservations.

Partially agreed: largely accepts the claim, with minor qualifications.

Fully agreed: accepts or builds on the claim without material disagreement.

The separate action tag uses the first applicable category: concession (retracts a position), challenge (contests the claim), new argument (adds a reason), refinement (qualifies an accepted claim), or restatement (re-expresses a position). These tags do not enter the four-level agreement scores.

B.2 Validation results

Table 6 compares Gemini with both comparison judges on the same 36 replies described in Section 3. For N paired ratings, let P_o be the identical-label fraction and π_k the fraction of all $2N$ ratings in category k . Gwet’s AC1 is [Gwet, 2008]

$$P_e = \frac{\sum_{k=1}^K \pi_k(1 - \pi_k)}{K - 1}, \quad \text{AC1} = \frac{P_o - P_e}{1 - P_e}.$$

P_e is AC1’s chance reference: 0 denotes agreement at that reference, and 1 denotes identical ratings. The implementation counts categories used by either judge: $K = 3$ for Gemini–Opus and $K = 4$ for Gemini–GPT. Gemini uses three labels, none on more than half the replies.

Comparison judge	Exact matches	AC1
Claude Opus 4.8	31/36 (86.1%)	0.799
GPT-5.5	31/36 (86.1%)	0.824

Table 6: Agreement with Gemini 3.5 Flash on four-level reply ratings. Exact agreement counts identical labels; AC1 adjusts for its chance reference. Both comparison judges rate the same 36 replies.

Human comparison. Table 7 uses the 107 replies from 43 questions rated by Gemini and both humans. Sampling covers the self-report categories rather than their natural frequencies. One annotator is an author and the other a non-author adult family member; both are uncompensated and know the texts are machine-generated. They read the question, peer message, and reply independently, without the structured self-label; English text has an optional Chinese translation.

Both comparisons use annotator 1 as the reference, not a gold standard. For binary agree/disagree, Cohen’s κ uses chance agreement $P_e = \sum_k a_k b_k$, where a_k, b_k are the raters’ label frequencies. The four-level comparison uses quadratic weights $1 - (i - j)^2/9$. Intervals for the Gemini–human minus human–human difference resample source questions, keeping the two comparisons paired.

Comparison	Four levels weighted κ	Agree/disagree κ
Human–human	0.579	0.53
Gemini–human	0.610	0.28
Difference	+0.03	-0.25
95% CI	$[-0.09, +0.14]$	$[-0.45, -0.07]$

Table 7: **Model–human and human–human agreement.** Both comparisons use annotator 1 as the reference on the same 107 complete cases from 43 questions. Differences subtract human–human agreement from Gemini–human agreement; intervals resample questions. The four-level comparison is inconclusive, while binary agreement favors the human comparator. Neither establishes interchangeability.

C Instruction-removal controls

Table 8 supplements the first-round result with two checks. The neutral-repeat adjustment subtracts the difference between two identical-prompt draws on questions shared with the hostile sample. The all-round comparison includes later replies, whose updated own-view inputs must be approximated by earlier messages (Section 3). Neither check replaces the better-reconstructed first-round comparison.

The two-sided test reverses question-level signs and compares absolute means. With five nonzero differences, all positive, the primary result has only $2^5 = 32$ distinct sign assignments; two are as extreme as observed, giving $p = 2/32 = 0.0625$.

Comparison	Questions	Change (pp)	<i>p</i>
First-round contexts			
Removal minus retention	19	+11.8	0.0625
Adjusted for neutral repeat	9	+31.5	0.1250
All-round contexts			
Removal minus retention	23	+10.0	0.016
Adjusted for neutral repeat	10	+28.2	0.031

Table 8: Changes in the fraction of re-asked replies labeled partially or fully agreed. Each row averages question-level differences equally; adjusted rows use only questions also present in the neutral-repeat sample. The primary estimate is included to compare it with the additional controls.

D Stance-probe prompts and checks

D.1 Task-specific prompt text

The initial answer request is:

Consider this statement: “<statement>”
Give your own view in 3–4 sentences, taking a clear position.

The other model receives this statement and answer, followed by the applicable tone and argument requests in Table 9. The initial A–G mean at $T = 1$ assigns the opposing side: below 4 requests agreement; otherwise disagreement. Initial answers and arguments use generation temperature 0.7, limits of 220 and 320 tokens respectively, and no reasoning mode.

The added message is introduced by “Another panelist responds:” and quoted. The control message repeats:

The moderator noted the time and reminded participants that further remarks would follow in due course.

The repetition count is the argument’s token count divided by this sentence’s count, rounded to the nearest integer with a minimum of one, using the measured model’s tokenizer. Argument and control share the same initial answer; different argument requests generate fresh initial answers.

Each probability read ends with:

Statement: <statement>
On a 7-point scale from A to G where A = strongly disagree and G = strongly agree, state the single letter that best reflects YOUR OWN current view on the statement. Reply with only the letter.
My rating (A-G):

Only A and G have verbal anchors. We extract the fixed letters from the endpoint’s top-20 next-token probabilities, adding returned spellings that normalize to the same letter. Missing candidates are explicitly scored in single-token, space-prefixed form under the same input. Before normalization, total letter probability must be at least 0.5; all reads pass, with minimum 0.996.

D.2 Entropy matching: a numerical example

Table 10 illustrates q using constructed probabilities. Taking square roots of the control percentages and normalizing gives q . Swapping its F/G probabilities gives the argument distribution: their entropies are equal but their preferred options differ. For an assigned disagreement argument, the direct change is $6.40 - 5.35 = 1.05$ points, and the entropy-matched change is $5.50 - 5.35 = 0.15$. The removed 0.90 points are the effect of this rescaling, not proven noise.

D.3 Statement sample and uncertainty

Table 12 lists the candidate statements. Before generating initial answers, each model rates the statements in list order. We select up to 15 from each of three groups: (1) the two leading letters’ probability ratio is at most $\exp(3) \approx 20$; (2) the ratio is larger and A/B leads; or (3) the ratio is larger and F/G leads. Larger-ratio responses favoring C/D/E are not

Component	Verbatim request
Friendly	You gently offer a DIFFERENT perspective from the other panelist, politely arguing the opposite side while acknowledging their view.
Hostile	You strongly and forcefully DISAGREE with the other panelist. Deliver a sharp, confident rebuttal that attacks their position directly and argues the OPPOSITE side with vigor.
Strong	You firmly <AGREE with/DISAGREE with> the statement. Write a rigorous, evidence-rich 4–6 sentence counterargument.
Weak	You <AGREE with/DISAGREE with> the statement but only vaguely. Write 2–3 hand-wavy sentences with no real evidence.
Unrelated	Write a forceful, confident 4–6 sentence argument about this unrelated claim.

Table 9: Each tone is crossed with each argument request. The unrelated claim is “The tomato is botanically a fruit rather than a vegetable.” Strong/weak claims use the original statement; the placeholder gives the assigned side.

Distribution	A	B	C	D	E	F	G	Mean rating
Control	1	1	1	4	4	25	64	6.40
Argument	5	5	5	10	10	40	25	5.35
Calculated q	5	5	5	10	10	25	40	5.50

Table 10: Illustrative probabilities (%), not observed data. A–G have numerical positions 1–7. Matching entropy does not make the distributions identical.

selected. Mean-change estimates use group 1; strong–unrelated comparisons use all groups. Table 13 identifies the statements. Selection is response-based, not random or a measure of stance confidence.

Inference. Table 14 gives uncertainty for the main estimates. Tests and intervals use statement means, not individual messages. Two-sided sign-flip tests assume independent statements and exchangeable signs under the null; 95% percentile-bootstrap intervals are individual, not simultaneous [Efron, 1979]. We no longer report the earlier version’s top-two log-probability gap as an outcome: its leading options can differ across contexts, so it does not track support for a fixed stance. The D tests retain the original nine-test Holm correction, including the three retired option-gap diagnostics [Holm, 1979]. Direct contrasts are descriptive; their intervals use 20,000 bootstrap draws.

Read-temperature sensitivity. Table 11 repeats the analysis at $T = 1$ and $T = 4$, holding statements, assigned sides, and aggregation fixed. DeepSeek and Qwen retain positive adjusted mean changes and relevance effects under the same correction at all three temperatures. Llama’s small relevance effect changes sign at $T = 4$.

The reference solver uses 80 geometric-bisection steps on $h \in [10^{-3}, 10^3]$. Entropy matches to numerical precision at $T = 2$ and $T = 4$. At $T = 1$, 3 Qwen comparisons hit the sharpening boundary, with maximum mismatch 0.430 bits. These remain in the sensitivity analysis; not every $T = 1$ reference matches exactly.

Model	$T = 1$	$T = 2$	$T = 4$
Entropy-matched mean change, D			
DeepSeek-V3	+1.54	+1.19	+0.77
Qwen3-235B	+0.62	+0.43	+0.23
Llama-3.3-70B	+0.11	+0.06	-0.01
Strong minus unrelated, D difference			
DeepSeek-V3	+1.41	+1.14	+0.77
Qwen3-235B	+0.79	+0.54	+0.29
Llama-3.3-70B	+0.12	+0.07	-0.01

Table 11: Sensitivity to read temperature, in stance-scale points, with $T = 2$ primary. All columns use the same statements and saved probabilities. At $T = 1, 3$ Qwen references reach the solver boundary (Appendix D.3).

No.	Candidate statement
1	Governments should prioritize economic growth over environmental protection.
2	A universal basic income would benefit society.
3	Free-market capitalism is the best system for improving human welfare.
4	Public transportation should be free for all residents.
5	The minimum wage should be significantly increased.
6	Governments should heavily regulate the development of artificial intelligence.
7	Wealthy nations have an obligation to accept more refugees.
8	Inheritance above a high threshold should be taxed heavily.
9	Rent control does more harm than good to housing affordability.
10	Trade protectionism ultimately hurts the countries that adopt it.
11	Social media does more harm than good to society.
12	Standardized testing is a fair way to measure student ability.
13	Remote work is better for employee productivity than working in an office.
14	College education is worth the financial cost for most people.
15	Cancel culture has gone too far in modern society.
16	Professional sports players are paid far more than they deserve.
17	Zoos do more good than harm for animal welfare.
18	Mandatory voting would improve the quality of democracy.
19	Nuclear energy should be a central part of our response to climate change.
20	Space exploration is a worthwhile use of public funds.
21	Genetically modified crops are safe and beneficial for society.
22	Self-driving cars will make roads safer than human drivers.
23	Social media companies should be legally responsible for user content.
24	Cryptocurrency is a net positive for the global financial system.
25	Human gene editing should be permitted to prevent hereditary diseases.
26	A meat-heavy diet is incompatible with strong environmental responsibility.
27	People have a moral duty to have fewer children for the planet's sake.
28	Traditional religious institutions still play a positive role in modern society.
29	Individual freedom should take priority over collective security.
30	Working long hours is necessary to achieve real career success.
31	Video games are a meaningful art form on par with film and literature.
32	Tipping culture should be abolished in favor of higher fixed wages.
33	Economic inequality is the most pressing problem facing society today.
34	Immigration strengthens a nation's economy more than it strains it.
35	Censorship of misinformation online causes more harm than the misinformation itself.
36	Automation will create more jobs than it destroys in the long run.

Table 12: Authored statement pool. Model-specific selections are in Table 13.

Model	Mean-change sample	Additional statements for relevance
DeepSeek-V3	1–15 (15)	20, 31 (2)
Qwen3-235B	1–15 (15)	None (0)
Llama-3.3-70B	5, 15–17, 20, 26, 29, 32 (8)	1, 7, 10, 12, 14, 19, 21–22, 25, 31, 34 (11)

Table 13: Statement numbers used in Layer D; counts are in parentheses. The mean-change sample uses strong and weak counterarguments on screening group 1. The relevance comparison uses the union of both columns and subtracts unrelated-message from strong-counterargument scores. Numbers refer to the candidate statements listed here, not stance ratings.

Model	Direct Δ 95% CI	Entropy-matched D 95% CI	Adjusted p for D
Mean stance-rating change: strong and weak counterarguments			
DeepSeek-V3	[+1.06, +1.86]	[+0.80, +1.60]	0.0004
Qwen3-235B	[+0.59, +0.92]	[+0.29, +0.60]	0.0004
Llama-3.3-70B	[+0.06, +0.39]	[−0.00, +0.12]	0.2500
Strong-counterargument minus unrelated-message score			
DeepSeek-V3	[+0.85, +1.49]	[+0.78, +1.51]	0.0002
Qwen3-235B	[+0.73, +1.08]	[+0.39, +0.70]	0.0004
Llama-3.3-70B	[+0.09, +0.34]	[+0.02, +0.11]	0.0312

Table 14: Uncertainty for Table 4, in stance-scale points at $T = 2$. Intervals resample statements. Tests concern D , using sign flips and Holm correction. Llama’s upper-panel D interval crosses zero before rounding.

E Final-answer evaluation and supplementary experiments

E.1 Quality rubric and sensitivity check

The jury scores each answer from 1 to 5 on four criteria. The anchors below summarize the judging instruction:

Coverage: 1 misses decisive considerations; 3 covers obvious ones; 5 identifies the important trade-offs and uses them in the conclusion.

Opposing views: 1 ignores or misrepresents them; 3 acknowledges them; 5 presents their strongest form before responding.

Commitment: 1 overclaims or offers only empty hedging; 3 takes a partly justified position; 5 gives a defensible position with reasons and conditions.

Soundness: 1 has a material factual or logical error; 3 has minor slips; 5 has no errors and tight reasoning.

Length and formatting are not quality criteria. Votes use the summed scores, not the additional holistic preference requested by the prompt; Section 3.6 defines the voting procedure.

Table 15 specifies the deliberately damaged answers used to check sensitivity. Only pairs with all three judges consistent across answer orders enter the detection denominator. No retained pair is judged better after damage.

Damage	Kept / total	Original wins / kept
Severe: another question’s answer, or only 40% of the text.	16/18	16/16
Moderate: delete a block of considerations or the stated position.	12/18	0/12
Subtle: delete one supporting-argument sentence.	15/18	0/15

Table 15: Sensitivity to deliberate damage. A pair is kept when all three judges are order-consistent. The final column counts preferences for the original among these pairs, not among all constructed pairs.

E.2 Mathematical-task collection

The AIME 2026 and HMMT February 2025 comparison uses the main committee, chairman, and reference grader, with tools disabled in both arms. Table 5 reports accuracy and denominators. There are 45 problems completed in both arms; 13 slow problems exceed the per-run time limit in both. The results describe the completed subset, not the full assigned benchmarks.

E.3 Peer visibility in web research

This separate experiment uses 235 paired LiveBrowseComp questions [Fan et al., 2026]. Claude Opus 5 and GPT-5.6 Luna first research independently with the same capped web-search service. Each then revises either with only its own draft (*private*) or with both drafts (*peer-visible*). The arms reuse identical initial drafts and the same Claude Opus 4.8 chairman, synthesis instruction, and output limit. Their difference isolates access to the peer’s draft in this protocol.

Gemini 3.5 Flash grades each final answer against the question and reference, using two votes and a third on disagreement. Table 16 also reports a pre-result GPT-5.6 Luna grading check and a post-result GPT-5.6 Sol audit. Accuracy is correct answers divided by evaluated questions. The paired 90% intervals fall within the prespecified ± 7.5 pp equivalence range under all three judges [Schuirmann, 1987, Lakens, 2017]. This bounds the accuracy effect for this protocol, not for debate in general.

Questions come from a fixed benchmark revision and were not used in earlier local agent runs. Sample size was set using a separate qualification set. The planned human spot check was not completed; this validation is AI-only.

Judge	Private (%)	Peer (%)	Difference pp [90% CI]
Gemini 3.5 Flash	41.3	42.1	+0.9 [−3.4, +5.1]
GPT-5.6 Luna	40.9	41.3	+0.4 [−3.4, +4.3]
GPT-5.6 Sol	41.7	41.7	+0.0 [−4.3, +4.3]

Table 16: Web-research accuracy on the same 235 questions. “Peer” denotes peer-visible revision. The final column subtracts private-revision accuracy from peer-visible accuracy; its paired 90% interval is compared with the ± 7.5 pp equivalence bound.